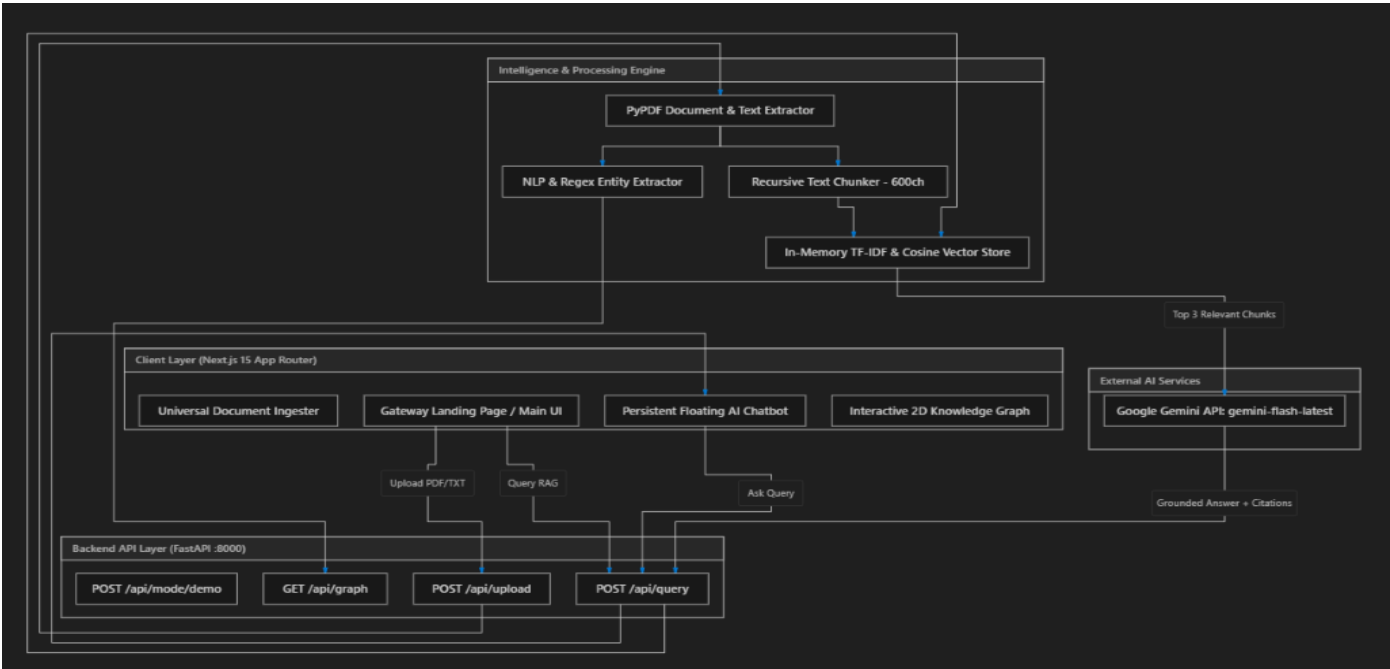


AURA-8 Architecture Diagram & Technical Specification

Economic Times AI Hackathon 2.0 | Problem Statement 8: AI for Industrial Knowledge Intelligence

1. High-Level System Architecture Diagram

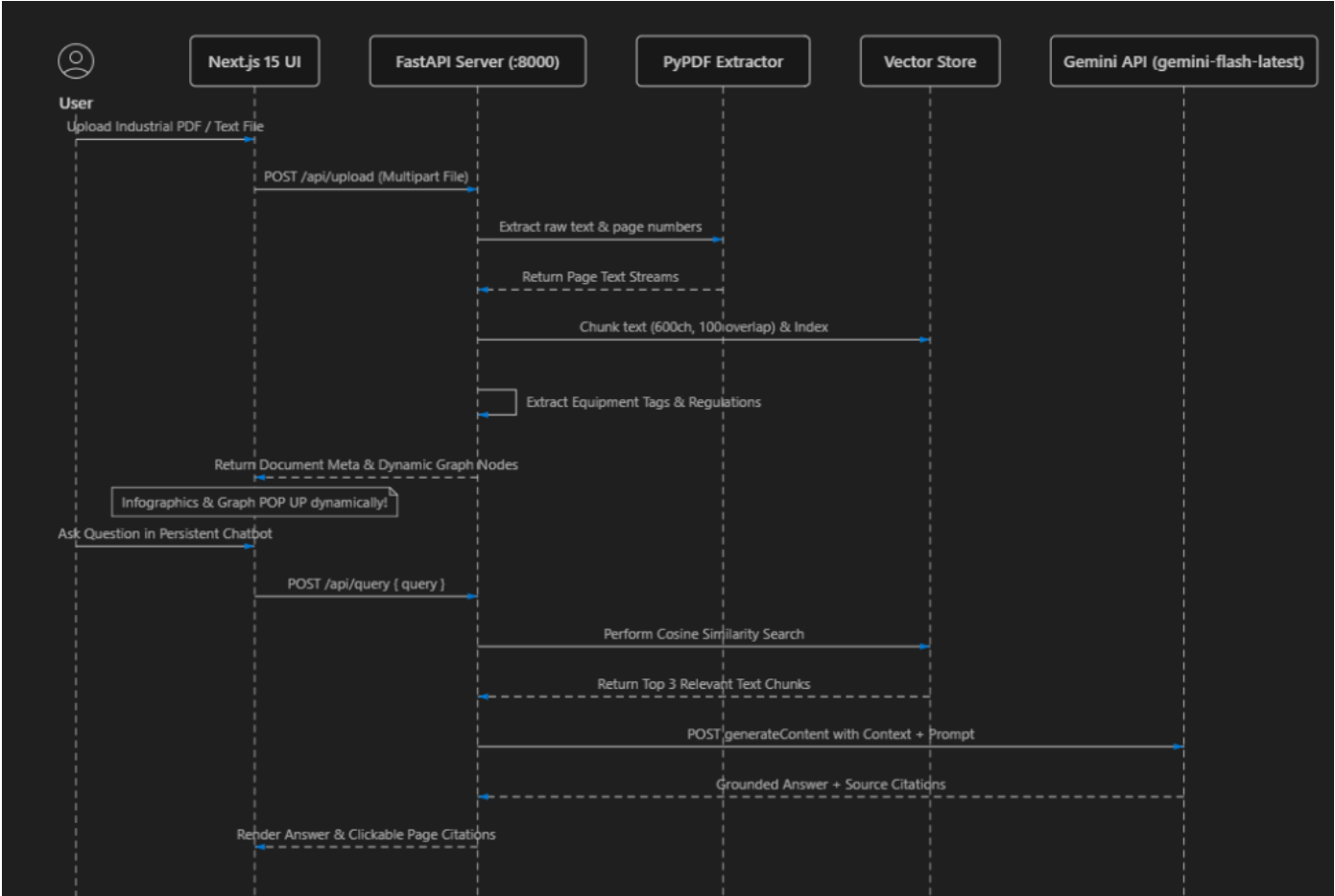
The system is structured across 4 decoupled architectural layers (Client UI → API Gateway → Intelligence Engine → Google Gemini API):



CLIENT LAYER (Next.js 15 App Router) - Gateway Landing Page (Demo vs Clean Start) - Persistent Floating AI Chatbot Drawer - Interactive 2D HTML5 Canvas Knowledge Graph - Universal Document Ingestor Dropzone	API ORCHESTRATION LAYER - POST /api/upload (PDF stream parsing) - POST /api/query (Gemini RAG call) - GET /api/graph (Knowledge Graph nodes) - POST /api/mode/demo (Sample hydrocracker data)
INTELLIGENCE & VECTOR ENGINE - pdf-parse & PyPDF Text Extractor - Recursive Text Chunker (600ch, 100 overlap) - TF-IDF & Cosine Similarity Vector Index - RegEx & NLP Entity Extractor (Tags P-101A, V-301)	EXTERNAL AI LLM SERVICE - Google Gemini API (gemini-flash-latest) - Endpoint: https://generativelanguage.googleapis.com/ - Header: X-goog-api-key - Output: Grounded Q&A; with page citations

2. Ingestion & RAG Execution Sequence Diagram

Detailed step-by-step sequence diagram for document upload and grounded RAG query processing:



Step #	Sequence Event	Source → Destination	Description & Payload
1	PDF Document Upload	User → Next.js UI	User drags PDF into dropzone or uploads via file selector.
2	Multipart POST Request	Next.js UI → /api/upload	Posts file FormData to same-origin relative API route.
3	Stream Decompression	/api/upload → pdf-parse	Decompresses /FlateDecode PDF binary stream into clean text.
4	Chunking & Vector Index	/api/upload → Vector Store	Splits text into 600-character overlapping chunks & indexes TF-IDF.
5	Entity & Tag Extraction	/api/upload → Entity Parser	Extracts equipment tags (P-101A) and statutory standards (OISD-137).
6	Graph Population	/api/upload → Knowledge Graph	Populates 2D Knowledge Graph nodes & unlocks dashboard infographics!
7	RAG User Query	User → Persistent Chatbot	User asks technical question in persistent AI chatbot drawer.
8	Cosine Match Search	/api/query → Vector Store	Retrieves top 3 relevant text chunks matching query term frequencies.
9	Gemini LLM Call	/api/query → Gemini API	Posts generateContent prompt with X-goog-api-key header.
10	Grounded Answer Render	Gemini API → Chatbot UI	Renders grounded response with clickable page-level text citations.

3. Technology Stack & Component Responsibilities

Decoupled architecture component mapping across frontend, backend, vector store, and LLM services:

Subsystem	Technology Stack	Core Responsibility
Frontend UI	Next.js 15, React 19, Tailwind CSS	Gateway landing page, persistent floating chatbot, interactive Canvas 2D graph, dark industrial design system.
RAG Backend	Next.js API Routes & Python FastAPI	Same-origin /api/upload and /api/query endpoints handling multipart requests with zero CORS issues.
LLM Engine	Google Gemini (gemini-flash-latest)	Synthesizing grounded responses, page citations, and executive summaries via X-goog-api-key header.
PDF Parser	pdf-parse (Node) & PyPDF (Python)	Decompressing PDF binary streams, extracting equipment tags (P-101A) and statutory OISD/Factory Act codes.